Abhinav Karnatak

abhinavkarnatak2004@gmail.com • +91-7310988194 • linkedin.com/in/abhinavkarnatak
github.com/abhinavkarnatak-dev • abhinavkarnatak.com

Software engineer building **full-stack applications** and **AI agents.** Built and deployed systems involving **secure code execution, GitHub automation, real-time communication, payments, and cloud infrastructure**.

SKILLS

AI Engineering: Agent Workflows, LangGraph, LangChain, RAG, Tool Calling, LLM API Integration, Prompt Engineering, AI Observability

Languages: Python, TypeScript, JavaScript, C++

Backend & Integrations: Node.js, Express.js, FastAPI, Pydantic, REST APIs, GitHub Apps, OAuth 2.0, JWT, E2B, WebSockets, Socket.io, RabbitMQ, Redis, Razorpay Payment Gateway

Frontend Technologies: React.js, Next.js, React Native, Tailwind CSS, Framer Motion

Databases: MongoDB, PostgreSQL, Drizzle ORM, Prisma

Cloud & DevOps: Docker, AWS EC2, AWS S3, Cloudflare R2, Nginx

Observability: PostHog

PROJECTS

Nimbus: Cloud Development Agent

- **Built a cloud development agent** that connects to GitHub repositories, understands codebase context, researches the web, and turns natural-language tasks into **verified software changes**.
- **Implemented sandboxed execution** for file inspection, code changes, command execution, artifact generation, Mermaid diagrams, and validation in an **isolated environment**.
- **Automated the GitHub workflow** with pause-and-resume sessions and PR creation, updates, merges, and closures, reducing the manual steps from **task to pull request**.

WakeOps: Automated Incident Response Platform

- **Built a full-stack incident response platform** using Next.js, Node.js, PostgreSQL, and Prisma that converts Grafana alerts into **tracked incidents with service and on-call context**.
- **Orchestrated automated incident calls** using Temporal and Twilio, with retries for unanswered calls, senior-engineer escalation, acknowledgements, and call history.
- **Decoupled email and Slack notifications** using RabbitMQ, allowing notification channels to run independently without **blocking the call escalation flow**.

YapIt: Real-Time Microservices Chat Application

- **Built a real-time chat application** using Next.js, Node.js, Socket.io, and MongoDB with **OTP authentication and WebSocket communication**.
- **Architected User, Chat, and Mail microservices**, using RabbitMQ for **asynchronous email delivery and account-cleanup tasks**.
- **Containerized and deployed the services on AWS EC2** using Docker Compose and Nginx, with S3 for image storage and Redis for **temporary OTP management**.

WORK EXPERIENCE

Software Engineer — **Sep 2025 - May 2026**
Capgemini — **Mumbai, Maharashtra (IN)**

- Tracked backend service and database health for a financial services platform, monitoring **CPU, memory, availability, and database load** through production dashboards.
- Analyzed **resource spikes and service degradation** using monitoring metrics, and coordinated with **cloud and database teams** when intervention was required.
- Maintained **production visibility across 20+ services**, enabling faster detection and response to **performance and availability issues**.

Backend Developer — **Oct 2024 - Dec 2024**
AdaptSense — **Dehradun, Uttarakhand (IN)**

- Developed a **React Native mobile app** using **Gemini APIs** to support **4 input modes** and **4 output formats** for accessibility-focused communication.
- Integrated **10+ input-output conversion combinations** in a single application, reducing the need for separate accessibility tools.

Data Engineer Intern — **Jun 2024 - Jul 2024**
HERE Technologies — **Remote**

- Developed **Python and Scrapy scripts** to collect retail store location data from multiple websites, handling different **HTML structures and page layouts**.
- Automated recurring data collection using **Scrapy spiders**, reducing **manual data collection effort by ~70%**.

EDUCATION

B.Tech Computer Science and Engineering — **Aug 2021 - May 2025**
University Of Petroleum and Energy Studies

Served as **Student Placement Representative (SPR)** for a batch of **100+ students**, coordinating communication with the placement cell and ensuring timely updates during campus recruitment drives.